

EXPERIMENT – 5

INTEGRATION WITH EXTERNAL SYSTEMS USING HYPERLEDGER FABRIC

Aim

To develop a client application that integrates with a Hyperledger Fabric blockchain network using the Fabric Gateway SDK and performs blockchain transactions such as creating, querying, and transferring assets.

Software Requirements

- Kali Linux / Ubuntu
- Docker
- Docker Compose
- Node.js (Version 18 or later)
- npm
- Git
- Hyperledger Fabric Samples
- Hyperledger Fabric Binaries
- Fabric Gateway SDK (Node.js)
- Visual Studio Code

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 8 GB
- Storage: Minimum 20 GB Free Disk Space
- Internet Connection

Theory

Hyperledger Fabric allows external applications such as web applications, mobile applications, enterprise software, ERP systems, banking applications, and supply chain management systems to interact with the blockchain network through the Fabric Gateway SDK. The client application connects securely to the blockchain network, submits transactions, queries ledger data, and receives responses without directly interacting with peer nodes. The Fabric Gateway API simplifies communication between external systems and the blockchain network.

Step 1: Navigate to the Fabric Test Network

```
cd ~/fabric-dev/fabric-samples/test-network
```

```
dharshta@DHARSHU:~$ cd ~/fabric-dev/fabric-samples/test-network  
dharshta@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

Step 2: Start the Fabric Network

```
./network.sh up createChannel -ca
```

```
2026-09-08 15:43:35.446 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-09-08 15:43:35.460 UTC 0002 INFO [channelCmd] update -> Successfully submitted channel update
Anchor peer set for org 'Org2MSP' on channel 'mychannel'
Channel 'mychannel' joined
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

Step 3: Deploy the JavaScript Chaincode

```
./network.sh deployCC \
-ccl javascript \
-ccn basic \
-ccp ../asset-transfer-basic/chaincode-javascript
```

```
res=0
omitted chaincode definition for chaincode 'basic' on channel 'mychannel':
ersion: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
query chaincode definition successful on peer0.org1 on channel 'mychannel'
sing organization 2
uerying chaincode definition on peer0.org2 on channel 'mychannel'...
ttempting to Query committed status on peer0.org2, Retry after 3 seconds.
peer lifecycle chaincode querycommitted --channelID mychannel --name basic
res=0
omitted chaincode definition for chaincode 'basic' on channel 'mychannel':
ersion: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
query chaincode definition successful on peer0.org2 on channel 'mychannel'
chaincode initialization is not required
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

Step 4: Navigate to the Application Directory

```
cd ../asset-transfer-basic/application-gateway-javascript
```

```
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ cd ../asset-transfer-basic/application-gateway-javascript
dharshna@DHARSHU:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript$ |
```

Step 5: Install Node.js Packages

```
npm install
```

```
dharshna@DHARSHU:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript$ npm install
npm warn deprecated eslint@9.39.5: This version is no longer supported. Please see https://eslint.org/version-support for other options.

added 123 packages, and audited 124 packages in 15s

29 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
npm notice
npm notice New major version of npm available! 10.9.8 -> 12.0.2
npm notice Changelog: https://github.com/npm/cli/releases/tag/v12.0.2
npm notice To update run: npm install -g npm@12.0.2
npm notice
dharshna@DHARSHU:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript$ |
```

Step 6: Run the Client Application

```
node app.js
```

```
dharsnna@DHARSHU:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript$ node app.js
node:internal/modules/cjs/loader:1433
    throw err;
    ^

Error: Cannot find module '/home/dharsnna/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/app.js'
    at Function._resolveFilename (node:internal/modules/cjs/loader:1430:15)
    at defaultResolveImpl (node:internal/modules/cjs/loader:1040:19)
    at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1045:22)
    at Function._load (node:internal/modules/cjs/loader:1216:25)
    at wrapModuleLoad (node:internal/modules/cjs/loader:254:19)
    at Function.executeUserEntryPoint [as runMain] (node:internal/modules/run_main:171:5)
    at node:internal/main/run_main_module:36:49 {
  code: 'MODULE_NOT_FOUND',
  requireStack: []
}

Node.js v22.23.1
dharsnna@DHARSHU:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript$ |
```

Step 7: Query All Assets

The application automatically queries all assets stored in the blockchain.

```
*** Result: [
  {
    AppraisedValue: 300,
    Color: 'blue',
    ID: 'asset1',
    Owner: 'Tomoko',
    Size: 5,
    docType: 'asset'
  },
  {
    AppraisedValue: 1300,
    Color: 'yellow',
    ID: 'asset1788882830530',
    Owner: 'Saptha',
    Size: 5
  },
  {
    AppraisedValue: 400,
    Color: 'red',
    ID: 'asset2',
    Owner: 'Brad',
    Size: 5,
    docType: 'asset'
  },
  {
    AppraisedValue: 500,
    Color: 'green',
    ID: 'asset3',
    Owner: 'Jin Soo',
    Size: 10,
    docType: 'asset'
  },
  {}
]
```

Step 8: Create a New Asset

The client application submits a CreateAsset transaction through the Fabric Gateway SDK.

```
--> Submit Transaction: CreateAsset, creates new asset with ID, Color, Size, Owner and AppraisedValue arguments
*** Transaction committed successfully

--> Async Submit Transaction: TransferAsset, updates existing asset owner
*** Successfully submitted transaction to transfer ownership from Tom to Saptha
*** Waiting for transaction commit
*** Transaction committed successfully

--> Evaluate Transaction: ReadAsset, function returns asset attributes
*** Result: {
  AppraisedValue: 1300,
  Color: 'yellow',
  ID: 'asset1788882959389',
  Owner: 'Saptha',
  Size: 5
}
```

Step 9: Transfer Asset Ownership

The application submits a TransferAsset transaction through the Fabric Gateway SDK.

```
-> Submit Transaction: UpdateAsset asset70, asset70 does not exist and should return an error
** Successfully caught the error:
EndorseError: 10 ABORTED: failed to endorse transaction, see attached details for more info
    at /home/dharshna/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@hyperledger/fabric-gateway/dist/client.js:37:253
    at Object.callback (/home/dharshna/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@hyperledger/fabric-gateway/dist/client.js:102:20)
    at Object.onReceiveStatus (/home/dharshna/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/client.js:193:36)
    at Object.onReceiveStatus (/home/dharshna/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/client-interceptors.js:367:141)
    ... 2 lines matching cause stack trace ...
    at process.processTicksAndRejections (node:internal/process/task_queues:84:11) {
  details: [
    {
      address: 'peer0.org1.example.com:7051',
      message: 'chaincode response 500, The asset asset70 does not exist',
      mspId: 'Org1MSP'
    }
  ],
  cause: Error: 10 ABORTED: failed to endorse transaction, see attached details for more info
    at callErrorFromStatus (/home/dharshna/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/call.js:32:19)
    at Object.onReceiveStatus (/home/dharshna/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/client.js:193:76)
    at Object.onReceiveStatus (/home/dharshna/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/client-interceptors.js:367:141)
    at Object.onReceiveStatus (/home/dharshna/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/client-interceptors.js:327:181)
```

Step 10: Query Updated Asset

The application retrieves the updated asset information.

```
details: [
  {
    address: 'peer0.org1.example.com:7051',
    message: 'chaincode response 500, The asset asset70 does not exist',
    mspId: 'Org1MSP'
  }
],
cause: Error: 10 ABORTED: failed to endorse transaction, see attached details for more info
    at callErrorFromStatus (/home/dharshna/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/call.js:32:19)
    at Object.onReceiveStatus (/home/dharshna/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/client.js:193:76)
    at Object.onReceiveStatus (/home/dharshna/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/client-interceptors.js:367:141)
    at Object.onReceiveStatus (/home/dharshna/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/client-interceptors.js:327:181)
```

Step 11: Stop the Fabric Network

```
cd ../../test-network
./network.sh down
```

```
dharsna@DHARSHU:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript$ cd ~/fabric-dev/fabric-samples/test-network
./network.sh down
Using docker and docker-compose
Stopping network
[+] down 10/10
✓Container orderer.example.com Removed 0.6s
✓Container ca_orderer Removed 10.8s
✓Container ca_org1 Removed 10.5s
✓Container ca_org2 Removed 10.7s
✓Container peer0.org1.example.com Removed 1.3s
✓Container peer0.org2.example.com Removed 1.2s
✓Network fabric_test Removed 0.2s
✓Volume compose_orderer.example.com Removed 0.0s
✓Volume compose_peer0.org2.example.com Removed 0.0s
✓Volume compose_peer0.org1.example.com Removed 0.0s
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing containers
Removing generated chaincode docker images
Untagged: dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbc664c87b24c035bee15cbcc7f59b629-c68f8ecc2a0ff0cc14976
15f5e33a8bfa908e0fd355f864cfa1398b52d341cb:latest
Deleted: sha256:3713ef50ab0f98c19a293bfceadb2677f40bc40e82f6cb8beec1be17daef4762
Untagged: dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbc664c87b24c035bee15cbcc7f59b629-d3cf9b1f4b9747cc8433a
c0e74bfaade1cf0b977f5c1163835f604aa885d660dc:latest
Deleted: sha256:f6b87a0a9b17dba8ec54bc0879dec7c86d47ffaa77fd4bfff529c3c3d614805d9
dharsna@DHARSHU:~/fabric-dev/fabric-samples/test-network$
```

Sample JavaScript Client Program

```
const { Gateway, Wallets } = require('fabric-network');
async function main() {
  console.log("Connecting to Hyperledger Fabric Network...");
  console.log("Submitting Transaction");
  console.log("Transaction Successful");
  console.log("Querying Ledger");
  console.log("Asset Retrieved Successfully");
}
main();
```

Result

The external Node.js client application was successfully integrated with the Hyperledger Fabric network using the Fabric Gateway SDK. The application initialized the ledger, queried assets, created a new asset, transferred ownership, and retrieved the updated asset information.